

Genitive vs. PP Arguments in German NPs

Antonio Machicao y Priemer

Pauline Reiß

Giuseppe Varaschin

Abstract

In German, NP arguments of nominal heads can be marked either with genitive case or with the preposition *von*. Although these types of marking are truth-conditionally equivalent, they have distinct contextual distributions, with genitive-marked arguments being associated with speakers who present themselves socially as more educated than *von*-marked arguments. This paper has a threefold objective. First, we offer a lexicalist account of the *von* vs. genitive alternation for NP arguments in German, deriving the pool of possible variants. Second, we formulate an extension of the HPSG architecture that allows us to predict the differences in usage preferences in a way that captures a graded notion of register and contextual appropriateness – in the spirit of Varaschin, Machicao y Priemer & Lu (2025). Our extension crucially involves the statement of use-conditional constraints, which associate independently licensed linguistic objects to gradable social meanings. Finally, we propose a corpus methodology to empirically validate hypotheses about social meaning assignments via use-conditional constraints and the principle of social meaning composition, focusing on the interaction between the *von* vs. genitive alternation and *-ung* (‘-ment’) nominalizations.

1 Introduction

In German, NP arguments of nominal heads can be marked either with genitive case (GEN) or with a PP headed by the preposition *von* (*von*-PP), which assigns dative case to the embedded NP (Machicao y Priemer 2017; Machicao y Priemer & Müller 2021; Kopf & Bildhauer 2024).

- (1) die Behandlung { des Patienten / von dem Patienten }
the treatment the.GEN patient.GEN of the.DAT patient.DAT
‘the treatment of the patient’

While the GEN NP and the *von*-PP are truth-conditionally equivalent, they differ w.r.t. the contexts in which they usually appear. For instance, in more educated or more formal contexts, the GEN realisation of the argument is preferred over its realisation within a *von*-PP. In contrast, in neutral or colloquial contexts the *von*-PP is preferred.

Similarly, the contexts in which *ung*-nominalisations¹ occur tend to be associated with a higher degree of education and formality (cf. nominal vs. verbal style). In this paper, we focus on verbs nominalised with the *-ung* suffix, as these are cases in which the argument structure of the verbal base is inherited by the derived nominal stem. Moreover, using *-ung* nouns as a

¹The nominalisation suffix *-ung* in German is comparable with the suffix *-ment* in English. For further details on this kind of nominalisation, cf. Ehrich & Rapp (2000); Dölling (2015); Machicao y Priemer (2017).

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proxy for education/formality enables us to examine which form (i.e. GEN or *von*) shows a stronger correlation with *-ung* in the data.

The goals of this paper can be summarised as follows: (i) We propose an account of the grammatical conditions for the *von* vs. GEN alternation, which also captures the parasitic behaviour of the preposition. (ii) We show how the HPSG architecture can be used to model not only the grammatical properties of the construction, but its use conditions, thereby predicting the contexts in which the construction occurs. (iii) We outline a method for empirically validating the theoretical implications of our analysis by means of corpus data.

2 Describing the variation

Within the German verbal domain, NP arguments of verbal heads that are marked with structural case are realised with nominative (NOM) and accusative (ACC) case, cf. (2).

- (2) *Tim* behandelt *den* *Patienten*.
Tim.NOM treats the.ACC patient.ACC
‘Tim treats the patient.’

When the verbal base is nominalised (e.g. with the suffix *-ung*) its argument structure is inherited by the derived nominal (Bierwisch 1989; Grimshaw 1990; Machicao y Priemer & Müller 2021), cf. (2) vs. (3). According to the Case Principle (Przepiórkowski 1999: 65), the NP arguments of the head are realised with GEN in the nominal domain – that is, GEN functions as the realisation of structural case within the nominal domain.

- (3) *Tims* Behandlung *des* *Patienten*
Tim.GEN treatment the.GEN patient.GEN
‘Tim’s treatment of the patient’

When realised postnominally,² the inherited arguments with structural case (i.e. GEN) can also be realised with a PP headed by the preposition *von* ‘of’ (cf. Machicao y Priemer 2017; Kopf & Bildhauer 2024), cf. (4a) & (4b) vs. (3). In this case, *von* selects an NP realised with dative (DAT).

- (4) a. *die* Behandlung *vo-m* *Patienten*
the treatment of-the.DAT patient.DAT
‘the treatment of the patient’

²In this paper, we concentrate on the alternation of GEN vs. *von* PP in postnominal position. For an account of the prenominal vs. postnominal realisation of arguments, cf. Machicao y Priemer & Müller (2021).

- b. die Behandlung *von Tim*
 the treatment of Tim.DAT
 ‘the treatment of Tim’

As a result of this alternation – and assuming that selection is strictly local (cf. selectional localism, Pollard & Sag 1987; 1994; Sag 2007; 2012) – the NP *dem Patienten* receives DAT from the preposition *von*. However, since the θ role is determined and assigned by the head noun *Behandlung*, only the PP (and not the NP itself) can receive this θ role, cf. Fig. 1. Note that the interpretation of the NP (or *von* PP) arguments of the head noun parallels that of the corresponding arguments in the verbal domain, cf. (2). Within the nominal domain an additional *possessor* interpretation is also available, which we likewise treat as an argument of the head noun (based on distributional evidence, cf. Machicao y Priemer & Müller 2021). Classifying these elements as arguments of the head noun – rather than as modifiers (contra Kopf & Bildhauer 2024) or arguments of an external head (contra Kratzer 1996) is crucial for an accurate account of the phenomenon.

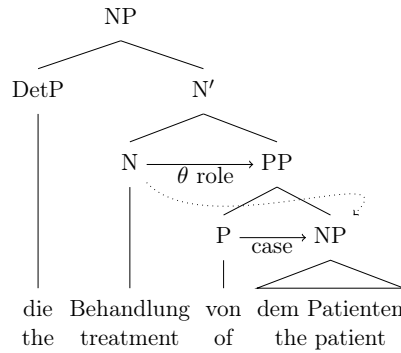


Figure 1: θ role and case assignment

That is, one puzzle we encounter in dealing with this structure is how θ role and case can be assigned by two different heads.

Furthermore, as pointed out in Kopf & Bildhauer (2024), not all *von* PPs undergo the alternation. For instance, some nouns – e.g. *Abhängigkeit* ‘dependence’ – select a *von* PP, cf. (5a). In this case, the *von* PP is selected also by the adjectival base *abhängig* ‘dependent’, cf. (5b), and is inherited by the nominalised head.³

- (5) a. die *Abhängigkeit* { von seinen Eltern / *seiner Eltern }
 the dependence on his parents his.GEN parents.GEN
 INTENDED: ‘the dependence *on* his parents’

³This is similar to the selection of an *of* PP by the verb *think*, which can also not alternate, cf. *I think of Tim* vs. *Tim’s thinking*.

- b. Der Student ist [AP von seinen Eltern *abhängig*].
 the student is on his parents dependent
 ‘The student depends on his parents.’

Another case in which the alternation does not take place is when the *von* PP is a true modifier of the noun (e.g. temporal or local), cf. (6). In this case, the preposition assigns case as well as a θ role to the NP it combines with – in comparison to the *von* used in the alternation.

- (6) der Blick { vo-m Turm / *des Turms }
 the view from-the tower the.GEN tower.GEN
 INTENDED: ‘the view from the tower’

For an adequate description of the alternation it is necessary to take into account that not all types of argument NPs can be realised marked with GEN case. Let us first examine the cases that allow both realisations, i.e. with GEN as well as with *von*. These cases include proper names, cf. (7a) & (8a), NPs with a definite determiner, cf. (7b) & (8b), and NPs with an indefinite determiner, cf. (7c) & (8c).

- | | |
|--|---|
| <p>(7) die Untersuchung ...
 the examination</p> <p>a. Tims
 Tim.GEN</p> <p>b. des Lehrers
 the.GEN teacher.GEN</p> <p>c. eines Lehrers
 a.GEN teacher.GEN</p> | <p>(8) die Untersuchung ...
 the examination</p> <p>a. von Tim
 of Tim.DAT</p> <p>b. vo-m Lehrer
 of-the.DAT teacher.DAT</p> <p>c. von einem Lehrer
 of a.DAT teacher.DAT</p> |
|--|---|

That is, a full NP can alternate, cf. (7) & (8), whereas an N' cannot serve as an argument either of the head noun (9a) or of the preposition (9b). This follows from independent reasons: since German singular bare nouns are not fully saturated phrases they cannot occur in argument positions.

- (9) a. *die Untersuchung *Lehrers*
 the examination teacher.SG.GEN
 b. *die Untersuchung *von Lehrer*
 the examination of teacher.SG.DAT

On the other hand, singular bare mass nouns (e.g. *Honig* ‘honey’) and plural count nouns (e.g. *Lehrer* ‘teachers’) are considered full NPs as well, as they can normally occupy argument positions without a determiner. However, they can appear only as arguments of the preposition *von*, cf. (11), and not as GEN marked arguments of the noun, cf. (10).

- | | |
|--|--|
| (10) die Untersuchung ...
the examination

a. * Honigs
honey.SG.GEN

b. * Lehrer
teacher.PL.GEN | (11) die Untersuchung ...
the examination

a. von Honig
of honey.SG.DAT

b. von Lehrern
of teacher.PL.DAT |
|--|--|

What is even more puzzling is that singular bare mass nouns and plural count nouns without a determiner can appear as GEN marked arguments of the noun when they are modified, cf. (12a) & (12b) vs. (10).⁴ This is unexpected, since the modifier does not turn a N into a NP. This can be seen with singular bare count nouns, which do not become fully saturated (i.e. NPs) by the addition of a modifier, cf. (9a) vs. (13)⁵ vs. (12b). Furthermore, given locality constraints, the selecting N-head should not be able to determine whether its argument is modified or not.

- | | |
|--|--|
| (12) die Untersuchung ...
the examination

a. <i>süßen</i> Honigs
sweet.SG.GEN honey.SG.GEN

b. <i>korrupter</i> Lehrer
corrupt.PL.GEN teacher.PL.GEN | (13) * die Untersuchung <i>korrupten</i> <i>Lehrers</i>
the examination corrupt.SG.GEN teacher.SG.GEN |
|--|--|

That is, we have to account for the fact that both the N-head and the preposition *von* combine with full NPs rather than with N', cf. (7) & (8) vs. (9). However, they differ w.r.t. their selectional restrictions: *von* can select “simple” or “complex” NPs, cf. (11), whereas the N-head selects only complex GEN NPs, cf. (12).

3 Grammatical constraints for alternation

Now that the issues related to the alternation have been identified, we can specify the grammatical constraints of the proposed approach. First, we need the lexical rule (LR) that derives *-ung* nouns and allows for the inheritance of arguments from the verbal to the nominal stem. We adopt the LR proposed in Machicao y Priemer & Müller (2021), which states that the arguments

⁴For the sake of completeness, it should be mentioned that *von* can combine with modified as well as with unmodified singular bare mass and plural count nouns.

⁵(13) is also not grammatical with the preposition *von*: **die Untersuchung von korruptem Lehrer* ‘the examination of corrupt teacher’.

with structural case and the PP arguments of the verbal base are inherited by the derived *-ung* noun, cf. (2) & (3).⁶ Following the Case Principle (Przepiórkowski 1999), arguments with structural case are realised with GEN in the nominal domain.

(14) LR for *-ung*-nominalisation (cf. Machicao y Priemer & Müller 2021)

$$\left[\begin{array}{l} \text{stem} \\ \text{PHON } \boxed{1} \\ \text{SYNSEM|LOC|CAT } \left[\begin{array}{ll} \text{HEAD} & \text{verb} \\ \text{ARG-ST } \boxed{2} & \text{list(str)} \oplus \text{list(lex)} \oplus \boxed{3} \text{ list(pp)} \end{array} \right] \end{array} \right] \mapsto \left[\begin{array}{l} \text{ung-n-stem} \\ \text{PHON } \boxed{1} \oplus \langle \text{ung} \rangle \\ \text{SYNSEM|LOC|CAT } \left[\begin{array}{ll} \text{HEAD} & \text{noun} \\ \text{ARG-ST } \boxed{2} \oplus \boxed{3} \end{array} \right] \end{array} \right]$$

For the mapping of elements from the ARG-ST list to the valence lists (e.g. COMPS), we assume argument structure mapping constraints (*as-mapping*) (cf. Manning & Sag 1998; Davis & Koenig 2000; Van Eynde 2015; Machicao y Priemer & Fritz-Huechante 2018; Machicao y Priemer & Müller 2021). To capture the fact that only “complex” NPs can be realised with GEN, cf. (12) vs. (10), we make use of the LEX(ICALICALITY) feature, introduced by Pollard & Sag (1987: 72–73).⁷ This feature distinguishes (roughly) between lexical and phrasal signs: LEX+ objects correspond “words”, while LEX− objects result from the combination of two objects licensed by a phrasal constraint. Accordingly, the “complex” NPs mentioned in Sec. 2 can now be characterised as the output of a phrasal combination, in the case of (12) specifically a *head-adjunct-phrase*. We implement this feature specification within our *as-mapping* constraint, such that only full NPs with the specification LEX− can be realised as complements of N-heads.⁸ With this constraint in place – that is independently needed not only for *-ung*-nouns but for all nouns – we can rule out N′ complements that are either LEX+ (9a) or LEX− (13), as well as full NPs that are not the result of a phrasal rule, i.e. LEX−, such as (10).⁹

(15) Constraint on ARG-ST to COMPS mapping

$$\left[\begin{array}{l} \text{n-as-mapping} \\ \text{HEAD } \text{noun} \\ \text{ARG-ST } \langle \dots, \boxed{1} \text{ NP[LEX } -], \dots \rangle \end{array} \right] \Rightarrow \left[\text{COMPS } \langle \dots, \boxed{1}, \dots \rangle \right]$$

⁶Following Przepiórkowski (1999), we divide the ARG-ST list into parametrised lists in order to inherit only the lists with arguments with structural case and PP arguments, but not the arguments with lexical case.

⁷See Arnold & Sadler (1992) and Deng et al. (2025) for further implementations of this feature for English and Mandarin Chinese.

⁸Recall that by virtue of the nominalisation LR in (14), only NPs with structural case are represented in the ARG-ST list of nouns.

⁹As mentioned above, the LEX+ vs. LEX− distinction corresponds “roughly” to the lexical–phrasal contrast. For instance, proper names such as *Tim* in (8a) can be classified as LEX−. This classification is also reflected in their behaviour w.r.t. modification and definiteness, and – depending on the analysis – they may even be derived via a phrasal constraint.

As noted w.r.t. Fig. (1), the preposition *von* assigns DAT (more specifically *lexical dative*) to the NP, while it is the N-head that assigns the θ role, resulting in a locality issue. Similar constructions are attested in other languages as well (for Dutch prepositions, see Van Eynde 2004; for *of* in English, see Van Eynde & Kim 2022; for *de* in Spanish, see Machicao y Priemer 2017; and for the *ass* camouflage construction, see Levine 2010). Following Levine (2010), we analyse this preposition as a *parasitic head*.¹⁰ Specifically, we posit a distinct subtype of the preposition *von* – different from the one used in modification, cf. (6), which assigns a temporal or locative θ role. This functional *von* (*von_f-prep*) acts as a case assigner, selecting a NP with lexical dative (*dem Patienten* in Fig. 2) and taking the INDEX value of its complement (*dem Patienten*) as its own.

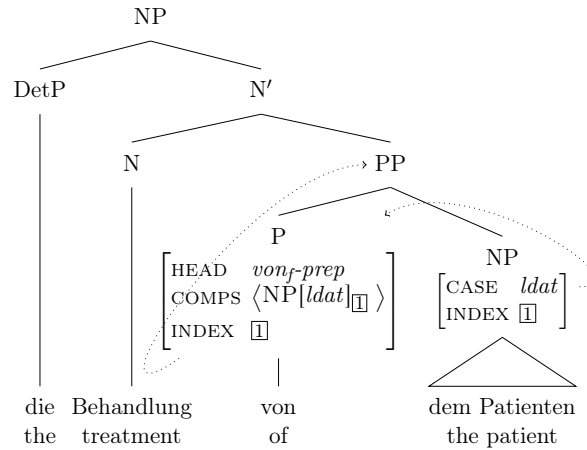


Figure 2: Parasitic head

In this way, the preposition can project its INDEX value to the maximal projection, which in turn receives a θ role from the N-head (*Behandlung* in Fig. 2). Consequently, the complement PP of the N-head (e.g. *Behandlung*) can bear either the agent or patient role, as illustrated in (4). The lexical entry for the preposition is provided in (16).¹¹

¹⁰Another possible analysis worth considering is the functor analysis of the preposition, following Van Eynde (2004); Van Eynde & Kim (2022). However, adopting this analysis would require more extensive changes to the overall system. Under such an approach, the resulting *von*-PP would be treated as an NP, which would in turn allow it to precede the N-head, yielding ungrammatical structures.

¹¹Take into account that this functional preposition is not only needed in this construction. This preposition fulfils the requirements to reintroduce the external argument in passive constructions in German, where the preposition assigns case but no θ role. For similar cases in Dutch, see Van Eynde 2004.

(16) Lexical entry for functional *von*

$$\begin{bmatrix} \text{PHON} & \langle \textit{von} \rangle \\ \text{HEAD} & \textit{von}_f\text{-prep} \\ \text{COMPS} & \langle \text{NP}[\textit{ldat}]_{\boxed{1}} \rangle \\ \text{INDEX} & \boxed{1} \end{bmatrix}$$

The final component of the grammatical analysis concerns the LR for responsible for the alternation. Although arguments realised with the preposition *von* are “more widespread” than those marked with GEN (cf. Sec. 4 & 5), the LR governing the alternation must take as its input an N-head that selects an NP with structural case. Recall that not all *von*-PPs participate in the alternation, cf. (6) & (5). Even in cases where *von_f* is used – this could be assumed for *Abhängigkeit* ‘dependence’ in (5) – the alternation is not possible. The crucial factor, then, is that a NP with structural case can be realised as a *von_f*-PP, whereas the reverse does not necessarily hold. We propose the LR in (17), which states that a NP with structural case can be realised as a PP with the *von_f* head.

(17) LR for GEN to *von_f*-PP alternation

$$\begin{bmatrix} n\text{-stem} \\ \text{CAT}[\text{ARG-ST} \langle \dots, \text{NP}[\textit{str}]_{\boxed{1}}, \dots \rangle] \end{bmatrix} \mapsto \begin{bmatrix} \text{CAT}[\text{ARG-ST} \langle \dots, \text{PP}[\textit{von}_f]_{\boxed{1}}, \dots \rangle] \end{bmatrix}$$

With these constraints in place, we can correctly predict distribution of the alternation in the cases illustrated in Sec. 2. All constraints used are in any case needed for independent reasons, i.e. in other constructions. We just needed to strengthen the restrictions on these constraints, for instance, restricting the *as-mapping* constraint for nouns w.r.t. the LEX value. Furthermore, we provide a more adequate description of the preposition *von_f*, which is not only needed in this construction, but can be used with other phenomena, e.g. passive. In the next section, we show how the grammatical and use conditions interact to predict in which contexts the alternation occurs.

4 Grammar and use conditions

In the previous section, we proposed a set of constraints that capture the range of realizations of NP arguments in German. However, as the brief remarks about the situational distribution of each variants suggest, speakers know more than the purely structural licensing conditions for NP arguments: they also know when (and by whom) each type of argument is preferably used. The question we address in this section is the following: how do we relate these two types of knowledge – knowledge of grammatical constraints and knowledge of usage preferences – in a way that allows our theory not only to predict the pool of possible variants, but also to say something about their relative frequencies in different linguistic and situational contexts?

Building on prior work (Asadpour et al. 2022; Machicao y Priemer, Müller, Schäfer & Bildhauer 2022; Varaschin, Machicao y Priemer & Lu 2025), we answer this question by formulating a theory of register in HPSG. The cornerstones of this theory are what we call **use-conditional constraints** (UCCs), which take the general form in (18):

(18) *description of linguistic structure $\mathcal{S} \Rightarrow$ description of $\mathcal{S}$'s context*

Formally, UCCs are just ordinary implicational constraints. On the left hand side, a UCC identifies a range of independently licensed structures by means of a partial description, which can be a type or a more complex feature description. The right hand side is a partial specification of the features of the context that are required to make the structure acceptable. In the case of HPSG, the context is represented as a feature of signs themselves, under the attribute `CONTEXT` (Green 1994). So UCCs, like other constraints, simply stipulate necessary conditions on the well-formedness of signs.

By using the HPSG formalism to state UCCs, we are hypothesizing that use-conditional knowledge is part of linguistic competence (Wilcock 1999; Paolillo 2000; Bender 2001; 2007; Asadpour et al. 2022, i.a.). It is useful, however, to distinguish between those aspects of linguistic competence that pertain to the well-formedness of `CAT`, `CONT` and their correspondence (i.e. the domain of narrow syntax and semantics) from those that pertain to contextual appropriateness. There is suggestive evidence that the former are acquired sooner and processed more rapidly than use-conditional information (Münster & Knoeferle 2018; Soto & Almeida 2021; Plesca et al. 2024, i.a.). This is ensured by keeping UCCs separate from the core grammatical constraints.¹² The set of UCCs is what Culy (1996) calls the ‘user’s manual’ of a language.

In the case of register variants, UCCs constraint contexts via the assignment of **social meanings** to linguistic structures (Burnett 2019; Beltrama 2020; Salmon 2022). We use this term in a broad sense to refer to any non-truth-conditional semantic content that ascribes some socially-relevant property to a context coordinate, for instance, the speaker (s_c) or the addressee (a_c). We are assuming, therefore, a multidimensional semantics in the tradition of Kaplan (1999), Potts (2005), McCready (2010) and Gutzmann (2015), whereby a sentence like (19a) would have two layers of meaning: a truth-conditional meaning (TC) and social meaning (SM), as in (19b-c).

¹²For the sole purpose of defining the models of a grammar, it would often not matter whether we encode constraints on `CONTEXT` via dedicated UCCs or introduce them as parts of other (independently required) constraints. For instance, suppose we want to say a form derived via a lexical rule carries a use-condition. We could either state the constraint on the `CONTEXT` as part of the statement of the lexical rule or define a separate UCC identifying the outputs of the lexical rule in the antecedent. We opt for the latter strategy to capture the idea that UCCs and core grammatical constraints have a different cognitive status. Note, however, that some UCCs – namely, those that identify antecedents that are not mentioned in other parts of the grammar – are not reducible to other constraints.

- (19) a. Die Behandlung des Patienten war gut.
 the treatment the.GEN patient.GEN was good
 b. TC: $\{w : \text{the treatment of the patient was good in } w\}$
 c. SM: $\{c : s_c \text{ is presenting as educated in } c\}$

As (19) makes clear the TCs an utterance u (TC_u) characterize a set of worlds. An utterance u is **true** in a specific world w iff $w \in TC_u$. In turn, the SM of an utterance u (SM_u) characterize a set of contexts by ascribing a property to one of its participants. An utterance u is **usable** in a particular context c iff $c \in SM_u$. (We refine this notion in gradient terms below.)

SMs can be distinguished from most instances of TC meanings on the basis of the following properties (see Varaschin, Machicao y Priemer & Lu 2025 for more discussion):

- (20) a. **Independence**: SMs are immune to the effect of TC operators (e.g. negation, modals, conditionals, interrogatives)
 b. **Unplugability**: SMs project over presuppositional plugs like propositional attitude predicates, they cannot be evaluated w.r.t. to any parameter other than the local context
 c. **Indexicality**: SMs predicate something of the present utterance situation or its participants (Potts 2007)
 d. **Gradability**: SMs hold of entities to different degrees

The first two properties are the most important ones, as they reflect the way SMs interact with TC content. To illustrate (20a), note that (21) conveys the genitive-induced SM we stated in (19c), in spite of the fact that the very proposition to which the genitive contributes its TC content is negated. (22) illustrates the related property in (20b): there is no reading in which the sentence conveys the SM that Lena (i.e. the attitude holder) is the one presenting as being educated. This SM projects to the speaker even under a reading where the existential presupposition associated with the definite *die Behandlung des Patienten* is not assumed to be satisfied in the world where the matrix clause is evaluated, but only in Lena's belief worlds. That is, even if we assume that no patient was treated in the actual world, the SM that comes from the genitive still projects. (23) illustrates the only context where SMs can be semantically embedded: direct quotation.

- (21) Die Behandlung des Patienten war **nicht** gut.
 the treatment the.GEN patient.GEN was not good
 (22) Lena **glaubt**, dass die Behandlung des Patienten gut war.
 Lena thinks that the treatment the.GEN patient.GEN good was
 (23) Lena sagte: „die Behandlung des Patienten war gut.“
 Lena said the treatment the.GEN patient.GEN was good

We propose to model properties (20a)-(20b) by placing SMs as the values of C(ONVENTIONAL)I(MPLICATION) attribute inside CONTEXT. In virtue of general linking constraints, TC operators (e.g. negation) and propositional attitudes only take their scopal arguments from elements under CONT. There is no TC operator that is a function over SM types. We model property (20c) by stating social-meaning relations in such a way that they always take C-INDEX value as one of their arguments. The property in (20d) is modeled by requiring SM relations to take a DEGREE argument (an interval from 0 to 1), as in Potts & Kawahara (2004) and McCready (2019). As in standard degree semantics, each point d in this interval stands for an equivalence class (Kennedy 2001. i.a.). In this case, ds are equivalence classes of contexts where the entity argument of the SM has the SM to an d -extent. This means that the wider the d range is the more contexts the SM is compatible with.

That said, we assume the following UCCs for the phenomena at hand:

- (24) **UCC** for NPs with arguments with structural case (i.e. genitive):

$$\left[\begin{array}{c} \text{HEAD} \quad \textit{noun} \\ \text{ARG-ST} \quad \langle \dots, \text{NP}[\textit{str}], \dots \rangle \end{array} \right] \Rightarrow \left[\text{CTXT} \left[\begin{array}{c} \text{C-INDS|SPEAKER} \quad \boxed{1} \\ \text{CI} \quad \left\langle \dots, \left[\begin{array}{c} \textit{educated} \\ \text{ARG1} \quad \boxed{1} \\ \text{DEG} \quad [.5, 1] \end{array} \right], \dots \right\rangle \end{array} \right] \right]$$

- (25) **UCC** for NPs with *von*-arguments:

$$\left[\begin{array}{c} \text{HEAD} \quad \textit{noun} \\ \text{ARG-ST} \quad \langle \dots, \text{PP}[\text{HEAD} \textit{von-prep}], \dots \rangle \end{array} \right] \Rightarrow \left[\text{CTXT} \left[\begin{array}{c} \text{C-INDS|SPEAKER} \quad \boxed{1} \\ \text{CI} \quad \left\langle \dots, \left[\begin{array}{c} \textit{educated} \\ \text{ARG1} \quad \boxed{1} \\ \text{DEG} \quad (0, .7] \end{array} \right], \dots \right\rangle \end{array} \right] \right]$$

- (26) **UCC** for *-ung* nouns:

$$\textit{ung-n-word} \Rightarrow \left[\text{CTXT} \left[\begin{array}{c} \text{C-INDS|SPEAKER} \quad \boxed{1} \\ \text{CI} \quad \left\langle \dots, \left[\begin{array}{c} \textit{educated} \\ \text{ARG1} \quad \boxed{1} \\ \text{DEG} \quad [.6, 1] \end{array} \right], \dots \right\rangle \end{array} \right] \right]$$

Each of these SMs assigns to a class of structures a SM that picks out different sets of contexts where the speaker is presenting as being *educated* to a specific degree. (24), for instance, assigns to *nouns* that have a argument marked with structural case (i.e. the genitive) a social meaning that represents a set of contexts where the speaker is presenting as being educated to mid-to-high extent. This captures the intuition that genitives sound more formal or educated in German. (26) assigns a similar SM (understood as a range of admissible contexts) to words that result from *ung*-derivations.

Of course this proposal is, as it stands, highly simplified. Forms subject to register variation are also typically not associated with a specific SM (e.g. *educated*), but with an indexical field of inferentially related SMs (Eckert 2008).

This can be modeled by associating these structures with an underspecified sort that includes *educated* as a subsort, so that SMs enforced by UCCs are only resolved to maximal sorts like *educated* in concrete communicative situations. For simplicity, we do not implement underspecification here, but a concrete worked-out proposal is developed in Varaschin et al. (2025).

The UCCs account for the SMs the grammar associated to individual pieces of structures. To calculate the SM of a full utterance, we need a principle of composition. A statement of the principle proposed by Varaschin, Machicao y Priemer & Lu (2025) is shown in (27). We say that two predications P_1 and P_2 are repeated predications *iff* they are of the same type and have the same ARG values but possibly different DEGR values.

(27) **Local CI Composition Principle**

For each phrase,

- a. If the CI values of its daughters do not have repeated predications, then the CI value of the phrase is the concatenation of the CI values of its daughters.
- b. If the CI values of its daughters have **repeated predications** $SM_1, \dots SM_n$ then the CI value of the phrase is the concatenation of the CI values of its daughters: (i) minus $\langle SM_1 \rangle, \dots \langle SM_n \rangle$; (ii) plus a list of predications of the same type and with the same ARG values as $SM_1, \dots SM_n$, but with a DEGR value consisting in the intersection between the DEGR values of $SM_1, \dots SM_n$.

In cases where repeated SMs do not intersect, the mother phrase will have a SM with an empty DEGREE value, which represents that the set of contexts where it is admissible is the empty set. This is arguably what happens with honorific mismatches (Kaur & Yamada 2022, i.a.). Crucially, however, having an empty DEGREE value does not mean that the structure is ungrammatical (i.e. not a model of the grammar), only that it is unusable, given that no utterance context will be an element of the empty set.

If they DEGREE values of repeated predications do intersect, they get progressively narrowed as one moves upwards in the tree. The core claim is that SM composition within a clause is monotonic: constraints on the context accumulate (i.e. each SM-bearing daughter *adds* restrictions) either via concatenation or via intersection of repeated SM predications. As we said, narrower DEGREE intervals represent a narrower set of admissible contexts.

In the case of German NPs, the core empirical prediction this approach makes, when combine with the UCCs in (24)-(26) is the following: *-ung* nouns and *von*-argument combinations should be more contextually restricted than combinations between *-ung* and genitives. Figures 3-4 illustrate the effects of SM composition in each of these cases. What is interesting here is that, due to the structures targeted by the UCCs, repeated SM predications have

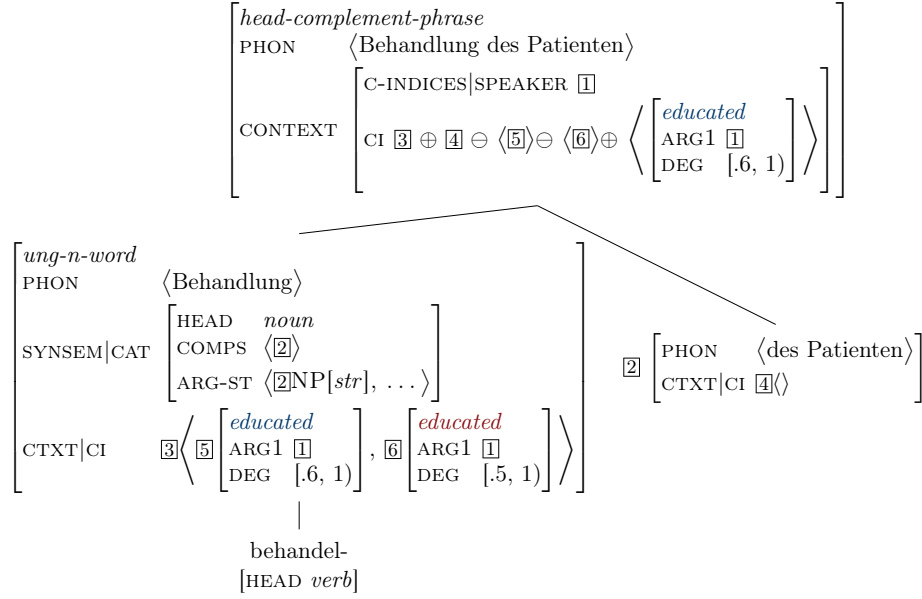


Figure 3: SM composition with *-ung* and genitive arguments

to be present at the lexical level. The composition principle in (27) applies when such words undergo phrasal combination.

If felicitous, the output of Local CI Composition (27) updates the global context – i.e. the context that pertains to the broader discourse beyond the sign (Paolillo 2000). Contextual update is subject to the following constraint:

(28) **Felicity constraint** (cf. McCready 2019: 31)

For every utterance u expressing a SM α , if the **prior global context** of u is specified as having a SM α' , where α and α' are repeated predications, then the DEGREE values of α and α' have to intersect.

The bigger the overlap between the SMs in the prior global context and those in u 's CONTEXT|CI value, the more appropriate u is with respect to the context. On this view, we can think of a register as a cluster of linguistic constraints whose associated models are required (by virtue of UCCs) to carry SMs that are appropriate in the same global contexts. Therefore, whether or not a form ‘belongs’ to a register R is a matter of degree, i.e. it depends on how much its SMs overlap with the contextual parameters associated with R .

5 Empirical Validation

In this chapter we argue that the proposed set of grammatical constraints and the proposed UCCs with respective DEG-values need to be empirically validated. Furthermore, we argue that all claims about register-sensitive

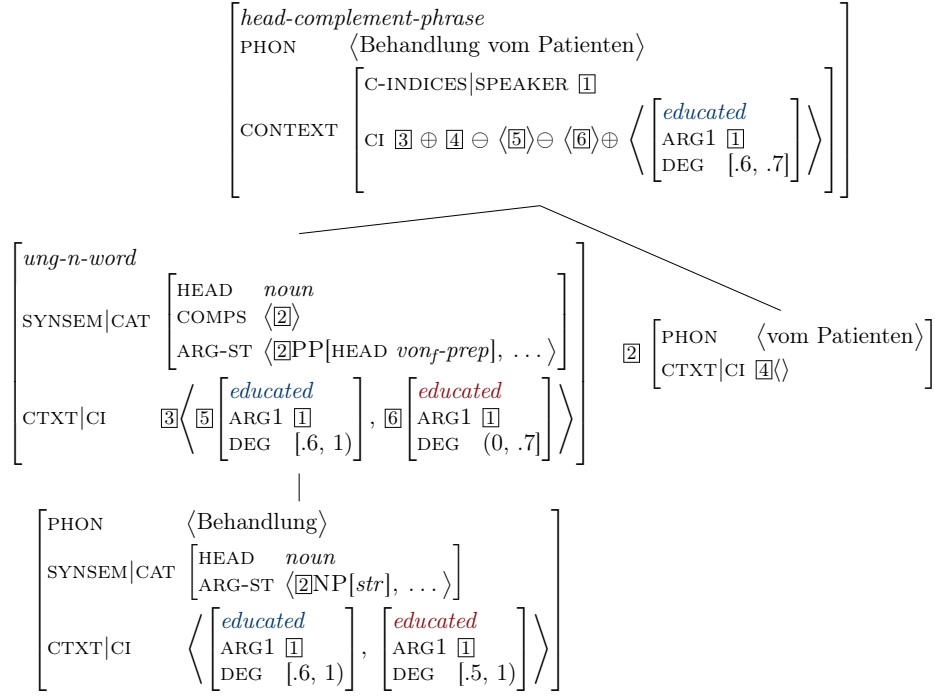


Figure 4: SM composition with with *-ung* and *von*-arguments

grammatical structures need to be supported by empirical data. In order to investigate the register sensitivity of genitives and *von*-phrases, we analyze corpus data from the PreCOXX25-LDA corpus (Schäfer, Bildhauer, Reiß, Pankratz & Müller 2025). Our goal is to test whether our theoretical predictions are empirically supported and, if necessary, make adjustments to our theoretical modeling. In this chapter we will outline our methodological approach, which should be seen as a proposal on how to motivate numerical values in UCCs and SMs.

The PreCOXX25-LDA corpus is a large-scale corpus containing 21,775,285 tokens across 2,475 documents. All documents show a wide variety of language representation, ranging from internet forums to sports reports and legal texts. We choose the corpus for its large number of tokens and linguistic heterogeneity. The corpus was build by Schäfer, Bildhauer, Reiß, Pankratz & Müller (2025) with a probabilistic approach based on Latent Dirichlet Allocation (LDA) (Blei 2012). LDA is a statistic model originally used for topic modeling, which posits that documents are mixtures of topics and topics are mixtures of words. As a result, documents are being assigned weights representing the probability that each document is associated with a specific topic. In the process of constructing the corpus, it was used to discover latent register dimensions to describe the probabilities of documents belonging to different potential registers (or *pregisters*), allowing the documents to be described as

a mixture of potential registers. In the experiment, 25 so-called preregisters were extracted. Each of these can be interpreted as a cluster of documents sharing similar linguistic features.

Since the analysis of register in language requires the analysis of situational and functional parameters, the clustering of texts into preregisters was completed by manual annotation. For the analysis, a subset of documents was extracted from the corpus and manually annotated by four human experts. The documents were classified according to situational and functional parameters: *Education*, *Interactivity*, *Proximity* and *Narrativity*, achieving a substantial to almost perfect inter-rater agreement across categories. Due to the combination of LDA-based analysis and human annotation, the corpus provides a robust empirical basis for analyzing register-sensitive grammatical structures and enables the nuanced analysis of register mixtures in texts.

In our analysis, we focus on the *Education* parameter due to its established association with nominal stylistics and genitive constructions (Biber 1988). In the manual annotation, *Education* was annotated using binary labels, following “whether a document requires an elevated educational background (EEB) on the side of the reader” (Schäfer, Bildhauer, Reiß, Pankratz & Müller 2025: 16) and achieved an overall substantial inter-rater agreement (Fleiss’s $\kappa = .67$).

For our analysis, we begin by ranking the preregisters by their *Education* annotation scores (drawn from the manual annotation), normalized on a continuum using min–max scaling. Preregisters with a larger proportion of documents annotated as *high education* receive a higher *Education* score whereas preregisters with a larger proportion of documents annotated as *low education* receive a lower score. These serve as proxies for the education parameter. We then compute whether the number of extracted genitives and *von*-phrases correlate with the assigned *Education*-scores. This methodology enables the comparison with the hypothesized DEG-values in the corresponding constructions.

For the analysis, we conduct a pre-study in order to pre-validate the hypothesis that genitives and *von*-phrases correlate in terms of their register usage. The pre-study is conducted without further annotation in order to ascertain the hypothesis’s plausibility and to obtain a preliminary insight. To visualize the relationships between genitives, *von*-phrases and the *Education*-scores, we create a heatmap (see Figure 5) to identify patterns, correlations, and anomalies within the dataset. By creating a heatmap (see Figure 5), the relationships between genitives, and *von*-phrases and the *Education*-scores can be visualized, allowing for the identification of patterns, correlations, and anomalies within the dataset.

The color gradient ranging from blue (-1.0) to red ($+1.0$) represents the strength of correlation, blue indicating a negative and red indicating a positive correlation. The heatmap reveals indications of weak to moderate correlations in the expected direction. Genitives and *von*-phrases show

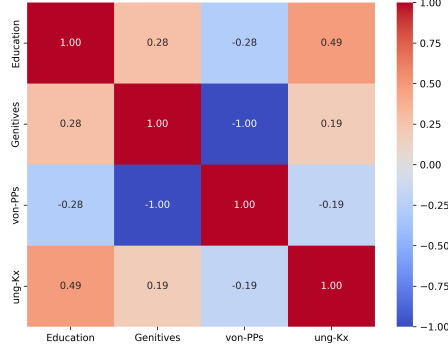


Figure 5: Correlation heatmap of all target variables, based on Spearman’s rank correlation coefficient.

a perfect negative correlation of -1 as the existence of one excludes the other. *ung*-nouns generally show a moderate positive correlation with the *Education* parameter. While *ung*-derived genitives show a weak positive correlation (0.28), *von*-phrases show a weak negative correlation (-0.28) with the *Education* parameter. This results enables us to infer that genitives and -*ung*-nouns trend positively with higher educational registers.

While the heatmap provides valuable insights, it is only a simplification of the studied structures. Still, these findings support the broader hypothesis of register-sensitive distribution patterns, as expected by our theoretical modeling. Therefore, we can deduce that the combinations of *von*-phrases with -*ung*-nouns should have small DEG values for their educated SM, rendering them more contextually restricted.

Nonetheless, validation via manual annotation is essential. As stated in chapter 2, the extent to which a speaker has the freedom to choose between the two constructions depends on the context. As a second step, we therefore conduct a manual annotation experiment to empirically validate the degree of optionality between genitives and *von*-phrases in our dataset. The study aims to determine whether speakers truly have the freedom to choose between the genitive and *von*-phrase in different contexts or if certain contexts favor one construction over the other. In order to maintain and verify the quality of annotations, the annotation was performed by two expert annotators. The quality of the annotation is calculated via inter-rater agreement (IRA) using Fleiss’s κ .

In total we conduct two annotation rounds, with the endeavour to finetune the annotation guidelines and to ensure sufficient inter-annotator agreement. The first round consists of $n=250$ items of two exemplary registers (10 and 24), resulting in $n=500$ in total. The registers were chosen based on their general high number of *ung*-derived nouns and their contrasting *Education*-scores.

Subsequent to the revision of discrepancies in the annotation guidelines, a second round was conducted to check the Operationalisation of the annotation and comprehensibility and clarity of the annotation guidelines. The second round consists of $n=250$ items from the same registers as well as an additional pre-register (10, 14 and 24), resulting in $n=750$ in total. The final IRA of Fleiss's $\kappa = .701$ indicates substantial agreement between the annotators, suggesting that the annotation guidelines were clear and consistently applied.

The annotation guidelines are based on the discussed literature in Sec. 2. First, we exclude all elements comprising proper names as attributes as they differ with regard to their syntactic properties and are, therefore, not suitable for comparison. In a second step, we filter the data based on optionality by excluding all items where speakers do not have choice over the construction. We start by excluding all items that are fixed constructions or idioms as in (29a). Additionally, all *von*-phrases are excluded if *von* has a temporal or local interpretation as in (29b) instead of an attributive one. If the *von*-phrase does not contain a determiner as in (29c) they are excluded, as genitive nouns are obligatorily marked with a determiner. This implies that no equivalent construction exists with a genitive. Lastly, all items are excluded if *von* is part of the argument structure of the verb as in (29d).

- (29) a. Tag der Arbeit
day the.GEN work.GEN
'Labour Day'
- b. der Blick vom Turm
the view from.the.DAT tower.DAT
'the view from the tower'
- c. Beförderung von Hunden
transport of dogs.PL.DAT
'transport of dogs'
- d. Ausschließung vom Wahlrecht
exclusion from.the.DAT voting.right.SG.DAT
'exclusion from voting rights'

The stepwise testing allowed us to establish the annotation guidelines and confirm their consistency, thus providing a solid foundation for a methodologically robust analysis.

Considering the two previously outlined annotation rounds it is evident, that establishing clear and consistent guidelines is essential prior to conducting a larger-scale empirical study. Despite the fact that the dataset is not sufficiently large and does not allow for a sophisticated statistical analysis, both annotation rounds still serve as a pre-study to pre-validate our hypothesis and to fine-tune the guidelines for the annotation. Only three registers were annotated to gain preliminary insights into the data. Preliminary findings

suggest that genitives and *-ung* nouns trend positively with higher education registers and that *von*-phrases show a slight negative correlation. While a larger statistical study is still pending, our methodology sets a strong foundation for validating the theoretical predictions regarding the encoding of register-sensitive grammatical structures, particularly in relation to education-based register variations. Once a larger study has been conducted, the data can be used to model the respective DEGREE values in the respective UCCs.

The presented methodological approach represents one of several possible methods for empirical validation in HPSG. It has potential applications for testing formal hypotheses about register and social meaning. Still, we wish to stress that numerical values in UCCs and SMs should not be arbitrary but empirically verified.

In this chapter, we presented a method combining probabilistic modeling (LDA) and human annotation for validating register-sensitive grammatical encoding. Future empirical work should expand the annotation to establish robust empirical intervals. While the analysis provides evidence supporting our hypothesis, further research is needed to refine the DEG values for our theoretical model.

6 Concluding remarks

In this paper, we proposed (i) structural constraints that predict when GEN-NPs and *von*-PPs **can** be used: (ii) UCCs that model when the two forms **would** be used, and (iii) a new method to empirically validate the predictions made by the model that results from (i)-(ii).

A crucial step in our methodological proposal was to take the LDA-induced registers ordered by their annotation-derived *education* scores as proxies for the *education* SM parameter itself. This assumes that how much a structure appears on a register assigned to a particular degree of education reflects how appropriate it is in a context where the speaker presents as having that particular degree of education.¹³ This appropriateness can then be compared to the hypothesized DEGR values we assign to the corresponding *educated* SM in UCCs like (26)–(25). If the empirically induced *education* intervals don’t match the hypothesized values, the latter can be revised accordingly. A further step is to develop some technique from automatically extracting SM DEGREE values from the distribution of different variants across different registers annotated for that SM.

On the theoretical side, our proposal says that genitive arguments are more frequent in more educated’ contexts because they are grammatically required

¹³This is a simplification (since many SM parameters are active in any situation), but it serves as a heuristically useful approximation for empirically validating hypothesized SMs. In this respect, this technique is similar to the use of written vs. spoken corpora as proxies for *formality*, as proposed in Sauerland (2022). The difference is that our association between corpora and situational parameters is independently established by annotators.

(by UCCs) to have SMs that largely overlap with these contexts. The higher prevalence of genitives in registers with high education scores was confirmed by our preliminary data analysis. According to the SM composition principle in (27), *-ung* derived Ns and *von*-PPs are predicted to have a restricted distribution because their hypothesized SMs define sets of admissible contexts with a narrow intersection. The preliminary data analysis we carried out favors this intersective approach to local SM composition. This goes against proposals like McCready’s (2019), who argues that DEGREE values of SMs of the same type are composed by calculating a numerical average.

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